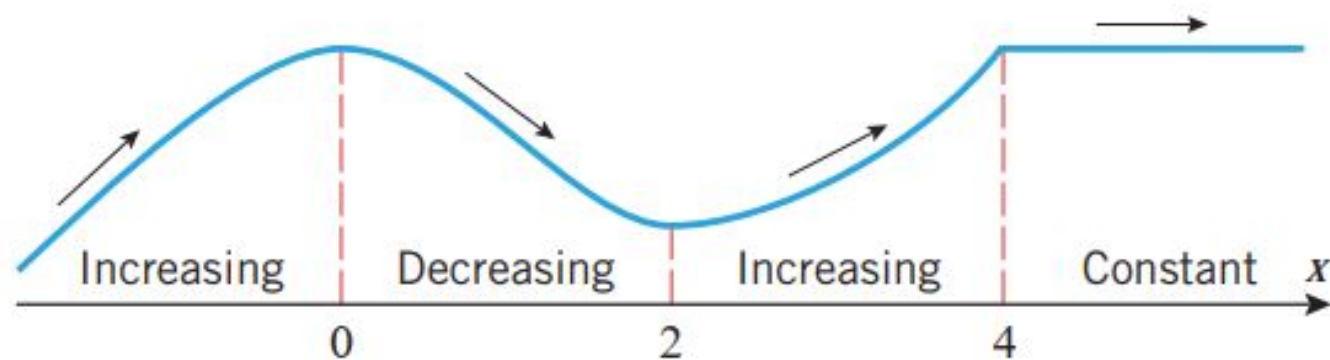


Increasing and decreasing interval

Prepared by Md Saif Kabir,
Lecturer, OAA

■ INCREASING AND DECREASING FUNCTIONS

The terms *increasing*, *decreasing*, and *constant* are used to describe the behavior of a function as we travel left to right along its graph. For example, the function graphed in Figure 4.1.1 can be described as increasing to the left of $x = 0$, decreasing from $x = 0$ to $x = 2$, increasing from $x = 2$ to $x = 4$, and constant to the right of $x = 4$.



► Figure 4.1.1

► **Example 5** Figure 4.1.10 shows the graph of the function $f(x) = x^3 - 3x^2 + 1$. Use the first and second derivatives of f to determine the intervals on which f is increasing, decreasing, concave up, and concave down. Locate all inflection points and confirm that your conclusions are consistent with the graph.

Solution. Calculating the first two derivatives of f we obtain

$$f'(x) = 3x^2 - 6x = 3x(x - 2)$$

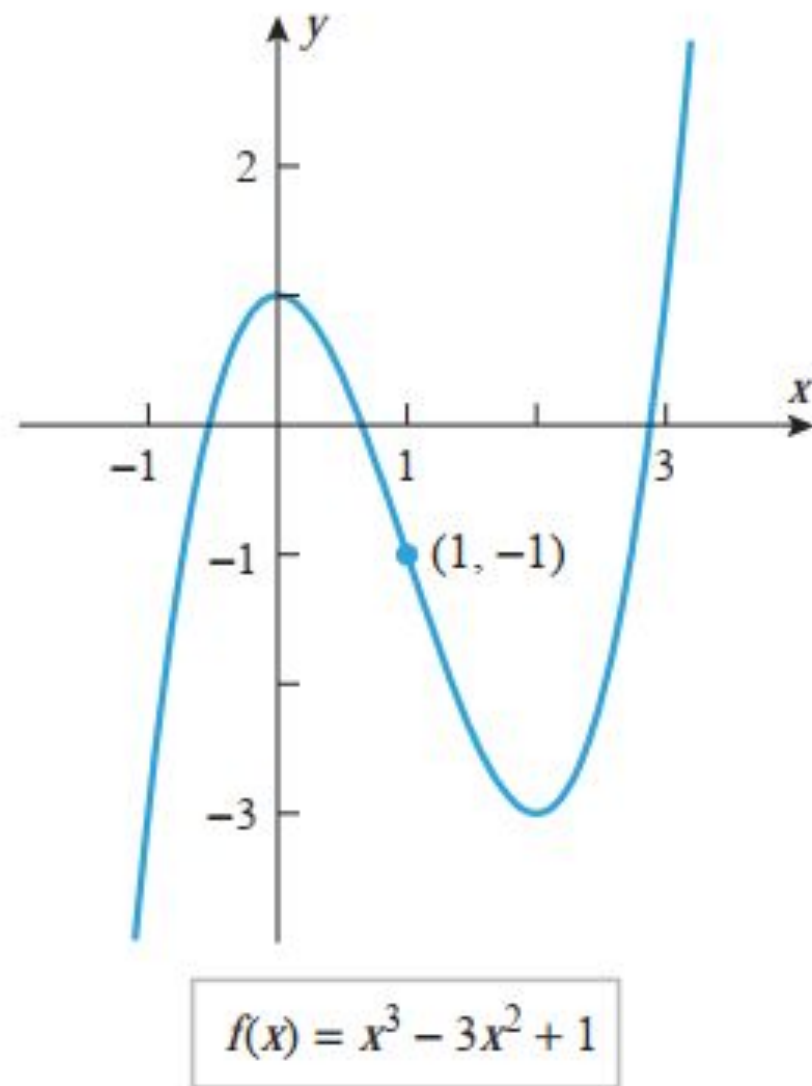
$$f''(x) = 6x - 6 = 6(x - 1)$$

The sign analysis of these derivatives is shown in the following tables:

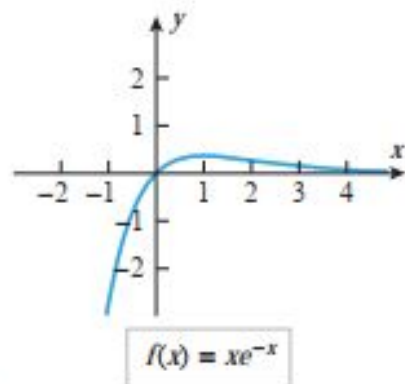
INTERVAL	$(3x)(x-2)$	$f'(x)$	CONCLUSION
$x < 0$	$(-)(-)$	$+$	f is increasing on $(-\infty, 0]$
$0 < x < 2$	$(+)(-)$	$-$	f is decreasing on $[0, 2]$
$x > 2$	$(+)(+)$	$+$	f is increasing on $[2, +\infty)$

INTERVAL	$6(x-1)$	$f''(x)$	CONCLUSION
$x < 1$	$(-)$	$-$	f is concave down on $(-\infty, 1)$
$x > 1$	$(+)$	$+$	f is concave up on $(1, +\infty)$

The second table shows that there is an inflection point at $x = 1$, since f changes from concave down to concave up at that point. The inflection point is $(1, f(1)) = (1, -1)$. All of these conclusions are consistent with the graph of f . ◀



► **Figure 4.1.10**



► Figure 4.1.11

► **Example 6** Figure 4.1.11 suggests that the function $f(x) = xe^{-x}$ has an inflection point but its exact location is not evident from the graph in this figure. Use the first and second derivatives of f to determine the intervals on which f is increasing, decreasing, concave up, and concave down. Locate all inflection points.

Solution. Calculating the first two derivatives of f we obtain (verify)

$$f'(x) = (1 - x)e^{-x}$$

$$f''(x) = (x - 2)e^{-x}$$

Keeping in mind that e^{-x} is positive for all x , the sign analysis of these derivatives is easily determined:

INTERVAL	$(1 - x)(e^{-x})$	$f'(x)$	CONCLUSION
$x < 1$	$(+)(+)$	$+$	f is increasing on $(-\infty, 1]$
$x > 1$	$(-)(+)$	$-$	f is decreasing on $[1, +\infty)$

INTERVAL	$(x - 2)(e^{-x})$	$f''(x)$	CONCLUSION
$x < 2$	$(-)(+)$	$-$	f is concave down on $(-\infty, 2)$
$x > 2$	$(+)(+)$	$+$	f is concave up on $(2, +\infty)$

The second table shows that there is an inflection point at $x = 2$, since f changes from concave down to concave up at that point. All of these conclusions are consistent with the graph of f . ◀

$$f(x) = x e^{-x}$$

$$f'(x) = x \frac{d}{dx} e^{-x} + e^{-x} \frac{d}{dx} x$$

$$= x \times e^{-x} \times \frac{d}{dx} (-x) + e^{-x} \cdot 1$$

$$= x e^{-x} \times -1 + e^{-x}$$

$$= e^{-x} - x e^{-x}$$

$$= (1-x) e^{-x}$$

$$f''(x) = 0$$

$$\Rightarrow (1-x) e^{-x} = 0$$

$$\Rightarrow x = 1, \quad e^{-x} = 0 \text{ doesn't exist}$$

$$f''(x) = (1-x) \frac{d}{dx} e^{-x} + e^{-x} \frac{d}{dx} (1-x)$$

$$= (1-x) \times e^{-x} \times \frac{d}{dx} (-x) + e^{-x} \times -1$$

$$= -e^{-x}(1-x) - e^{-x}$$

$$= -e^{-x} + x e^{-x} - e^{-x}$$

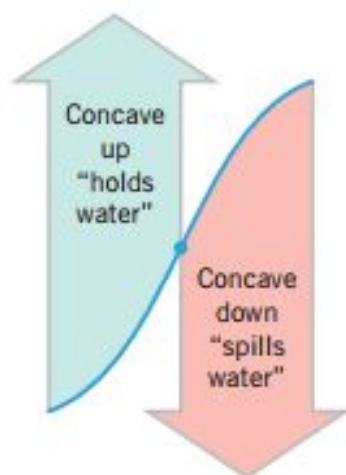
$$= x e^{-x} - 2 e^{-x}$$

$$= (x-2) e^{-x}$$

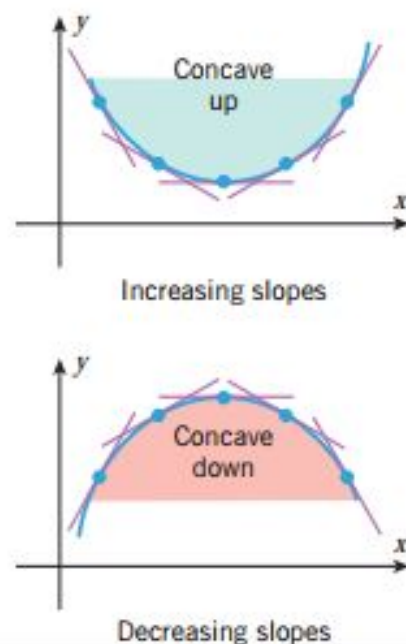
$$f''(x) = 0$$

$$\Rightarrow (x-2) e^{-x} = 0$$

$$\Rightarrow x = 2, \quad e^{-x} = 0 \text{ doesn't exist}$$



► Figure 4.1.7



■ CONCAVITY

Although the sign of the derivative of f reveals where the graph of f is increasing or decreasing, it does not reveal the direction of *curvature*. For example, the graph is increasing on both sides of the point in Figure 4.1.7, but on the left side it has an upward curvature (“holds water”) and on the right side it has a downward curvature (“spills water”). On intervals where the graph of f has upward curvature we say that f is *concave up*, and on intervals where the graph has downward curvature we say that f is *concave down*.

Figure 4.1.8 suggests two ways to characterize the concavity of a differentiable function f on an open interval:

- f is concave up on an open interval if its tangent lines have increasing slopes on that interval and is concave down if they have decreasing slopes.
- f is concave up on an open interval if its graph lies above its tangent lines on that interval and is concave down if it lies below its tangent lines.

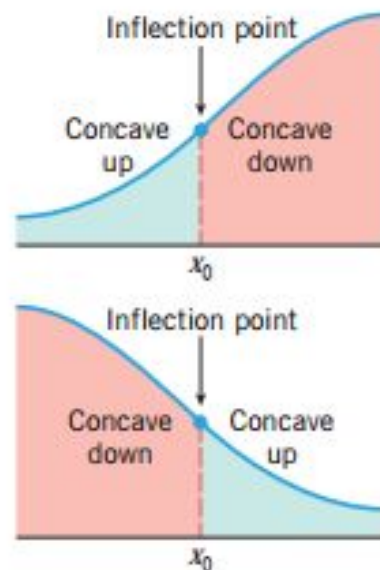
Our formal definition for “concave up” and “concave down” corresponds to the first of these characterizations.

4.1.3 DEFINITION If f is differentiable on an open interval, then f is said to be *concave up* on the open interval if f' is increasing on that interval, and f is said to be *concave down* on the open interval if f' is decreasing on that interval.

Since the slopes of the tangent lines to the graph of a differentiable function f are the values of its derivative f' , it follows from Theorem 4.1.2 (applied to f' rather than f) that f' will be increasing on intervals where f'' is positive and that f' will be decreasing on intervals where f'' is negative. Thus, we have the following theorem.

■ INFLECTION POINTS

We see from Example 4 and Figure 4.1.5 that the graph of $f(x) = x^3$ changes from concave down to concave up at $x = 0$. Points where a curve changes from concave up to concave down or vice versa are of special interest, so there is some terminology associated with them.



4.1.5 DEFINITION If f is continuous on an open interval containing a value x_0 , and if f changes the direction of its concavity at the point $(x_0, f(x_0))$, then we say that f has an *inflection point at x_0* , and we call the point $(x_0, f(x_0))$ on the graph of f an *inflection point* of f (Figure 4.1.9).

► **Example 5** Figure 4.1.10 shows the graph of the function $f(x) = x^3 - 3x^2 + 1$. Use the first and second derivatives of f to determine the intervals on which f is increasing, decreasing, concave up, and concave down. Locate all inflection points and confirm that your conclusions are consistent with the graph.

Solution Calculating the first two derivatives of f we obtain